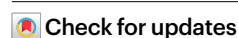


ImmunoStruct enables multimodal deep learning for immunogenicity prediction

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Epitope-based vaccines are promising therapeutic modalities for infectious diseases and cancer, but identifying immunogenic epitopes is challenging. Most prediction methods only use amino acid sequence information, and do not incorporate wide-scale structure data and biochemical properties across each peptide–major histocompatibility complex (MHC). We present ImmunoStruct, a deep learning model that integrates sequence, structural and biochemical information to predict multi-allele class I peptide–MHC immunogenicity. By leveraging a multimodal dataset of 26,049 peptide–MHCs, we demonstrate that ImmunoStruct improves immunogenicity prediction performance and interpretability beyond existing methods, across infectious disease epitopes and cancer neoepitopes. We further show strong alignment with in vitro assay results for a set of SARS-CoV-2 epitopes, as well as strong performance in peptide–MHC-based survival prediction for patients with cancer. Overall, this work also presents an architecture that incorporates equivariant graph processing and multimodal data integration for a long-standing challenge in immunotherapy.

Epitope-based vaccines are a type of targeted immunotherapy that uses short peptides to prime the immune system to attack infected or cancerous cells in the body. A peptide is considered to contain an immunogenic epitope if it can successfully invoke an immune response in the host, typically by activating T cells that target the same specific epitope. Epitope-based vaccines have shown promise in the clinic across cancer types such as melanoma, bladder carcinoma, glioblastoma, lung adenocarcinoma and pancreatic cancer^{1–3}. However, vaccine delivery constructs, such as lipid nanoparticles for mRNA vaccines, typically only hold up to 20–30 candidate peptides. Moreover, only a subset of patients respond to a limited fraction of these epitopes, and validating epitope immunogenicity remains resource intensive^{2–4}. Although patients with observed CD8⁺ T cell-mediated responses can exhibit encouraging clinical outcomes, improved tools are still needed for prioritizing and ranking epitope immunogenicity to enhance candidate selection. For instance, in previous analyses only 2–6% of

predicted mutation-derived peptides were shown to actually result in true patient-derived T cell immunogenicity^{4–7}. Furthermore, recent cancer neoantigen vaccine trials have predominantly only used transcript expression and affinity-based methods for filtering and identifying patient-specific epitopes, resulting in a fraction of true positive predictions⁸. Therefore, given the rarity of immunogenic epitopes and the paucity of reliable prediction methods, an important challenge remains in developing methods to distinguish immunogenic peptides from those that are not to enhance the design of epitope-based vaccines.

The surveillance of the human immune system relies on the presentation of antigen peptides on top of MHC proteins, which occurs via two distinct peptide presentation pathways, namely class I and class II. MHC class I proteins are found on most cells. These proteins present peptides to cytotoxic CD8⁺ T cells, while MHC class II proteins are found on antigen-presenting cells and interact with helper T cells^{9,10}. A key goal in epitope-based vaccine design is to

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induce more robust CD8⁺ T cell responses towards pathogenic cells, but pinpointing class I immunogenicity determinants is complex as it involves multiple cellular mechanisms. These include antigen processing, peptide–MHC binding, peptide–MHC presentation, peptide–MHC–T cell receptor (TCR) recognition and CD8⁺ T cell activation⁹. The same peptide can also show varying CD8⁺ T cell activation on the basis of its paired MHC allele, which vary substantially across the human population¹⁰. These variable peptide–MHC responses exist partly because structural differences in the peptide–MHC also influence binding cooperativity^{11,12}. Despite the development of numerous peptide–MHC deep learning models such as NetMHCpan-4.1¹³, MHCnuggets¹⁴, MHCflurry-2.0¹⁵ and BigMHC¹⁶, most still rely mainly on peptide and MHC sequence data without incorporating multi-allelic structural information and biochemical information⁷, which have proved to be critical for peptide–MHC–TCR synapse recognition and immunogenicity¹⁷. NeoPred¹⁸, a recent structure-based approach demonstrates the potential to incorporate structural information, but its scope remains limited because it requires mutant–wild-type pairs for immunogenicity prediction and therefore primarily focuses on cancer neoantigens.

To address these challenges, we present ImmunoStruct, a deep learning model that integrates peptide–MHC sequence, structure and biochemical properties to predict class I peptide–MHC immunogenicity. This method leverages these combined modalities for immunogenicity prediction across both infectious disease and cancer.

ImmunoStruct is a multimodal neural network architecture, aiming to effectively integrate sequence information, structural peptide–MHC information and biochemical property information across all samples to improve immunogenicity prediction. ImmunoStruct consists of a variational autoencoder (VAE) module that encodes a compressed representation of a peptide–MHC sequence, a graph transformer module that processes the graph constructed from both sequence and structure, and a multilayer perceptron (MLP) layer that encodes biochemical information. Furthermore, ImmunoStruct features a multimodal attention mechanism that enables an innovative layer of structural interpretability over its predictions. In addition, given the availability of both wild-type and mutant peptide sequences in the cancer neoantigen context, ImmunoStruct uses a contrastive loss approach that leverages both the wild-type peptide–MHC data to enforce mutant–wild-type pairs to be farther away in latent space if the mutant variant is immunogenic and vice versa. This helps the model distinguish whether a given mutation in a wild-type sequence would yield an immunogenic mutant peptide.

To train ImmunoStruct, we custom-curated a multi-allelic dataset of peptide–MHC sequences using the Immune Epitope Database (IEDB)¹⁹ and Cancer Epitope Database and Analysis Resource (CEDAR)²⁰ database: IEDB data were used for infectious diseases and CEDAR data were used for cancer neoepitopes. We then generated the peptide–MHC using our high-throughput AlphaFold2 pipeline²¹, in which each peptide–MHC pair was jointly folded. Finally, biochemical properties were computed using both sequences and structures, creating our final multimodal dataset, which in total contained the three distinct modalities.

ImmunoStruct was evaluated on held-out test data from both infectious disease and cancer from IEDB and CEDAR, respectively, showing consistently improved performance over current state-of-the-art models^{13–16,22,23}. Moreover, given the importance of experimental validation especially for deep learning approaches, we generated an *in vitro* assay using ELISpot interferon- γ (IFN γ) for a set of 19 peptide–MHCs. These peptides comprised various components of SARS-CoV-2 proteins with confirmed CD8⁺ T cell immunogenicities. We also validated ImmunoStruct in a cohort of survival for patients with cancer comprising a variety of different tumour types and immunotherapeutic treatments. Together, with these experimental validation steps, ImmunoStruct also demonstrated clinical translation potential.

Overall, the key contributions of ImmunoStruct include the following:

- (1) A multimodal neural network architecture capable of synergistically integrating three distinct data modalities for the immunogenicity prediction task.
- (2) Curation of a multimodal dataset enabling the use of structural, sequence and biochemical property information to improve prediction of peptide–MHC immunogenicity.
- (3) Interpretability of peptide–MHC predictions derived from attention-guided biostructural insights for immunogenic peptide–MHC samples.
- (4) A contrastive loss approach for the classification of cancer neoepitopes that forces wild-type and immunogenic mutant peptides to be embedded farther away in latent space.

Results

The ImmunoStruct model

ImmunoStruct integrates protein sequence, structure and biochemical information for peptide–MHC immunogenicity prediction (Fig. 1a). The model takes in three input modalities: (1) peptide–MHC sequence, (2) peptide–MHC three dimensional (3D) structure given by AlphaFold2 (Fig. 1b), and (3) biochemical properties (Fig. 1c). These three modalities are modelled using a VAE, a graph transformer network and a MLP that, respectively, encode the sequence, the structure–sequence combination and the biochemical properties. The multimodal embedding is then passed to a downstream MLP to produce a single scalar value for the prediction of immunogenicity. Technical details on data curation and processing are further described in the Methods.

Structure–sequence joint modelling. Peptide–MHC structures are first predicted from peptide and MHC sequences using a high-throughput AlphaFold2 pipeline where we jointly fold the peptide into the MHC protein, thereby forming the peptide–MHC complex for each pair. These peptide–MHC structures contain both structural and sequence amino acid information, which are then converted into graphs that contain biochemical bonds of the peptide–MHC as the edges, as well as amino acid sequence and 3D positional information as the node features (Fig. 1b). These graphs are then processed using a graph transformer module that also leverages $E(n)$ -equivariant graph neural network (EGNN)²⁴ layers that are equivariant with respect to 3D rotation, translation, reflection and permutation²⁵. A multihead self-attention mechanism enables it to capture spatial relationships and attend to learned pairwise interactions at all amino acid positions of the peptide–MHC^{26,27}. This results in 64-dimensional structural embeddings, which are then integrated with the information processed from the other two modalities (Fig. 1a). Since data are represented and processed as graphs, for simplicity, we will refer to this joint modelling as ‘Structure’ in ablation studies.

Biostructural interpretation module. To directly visualize the learned structural attention patterns for the immunogenicity of peptide–MHC, we implemented a structural attention visualization module. This enables interpretable overlays for individual amino acid residues, node-to-node amino acid interactions and global attention-guided biostructural interpretations of the peptide–MHC surface (Supplementary Fig. 1a,b). The module allows for fast extraction and processing of learned pairwise sequence–structure attention weights across all nodes of the peptide–MHC and can be easily integrated with the raw protein structure file of 3D atomic coordinates generated by AlphaFold2.

Sequence modelling. We also explicitly encode peptide–MHC sequences by a VAE framework that is further processed by a multihead self-attention layer²⁸. The full sequences are obtained by concatenating

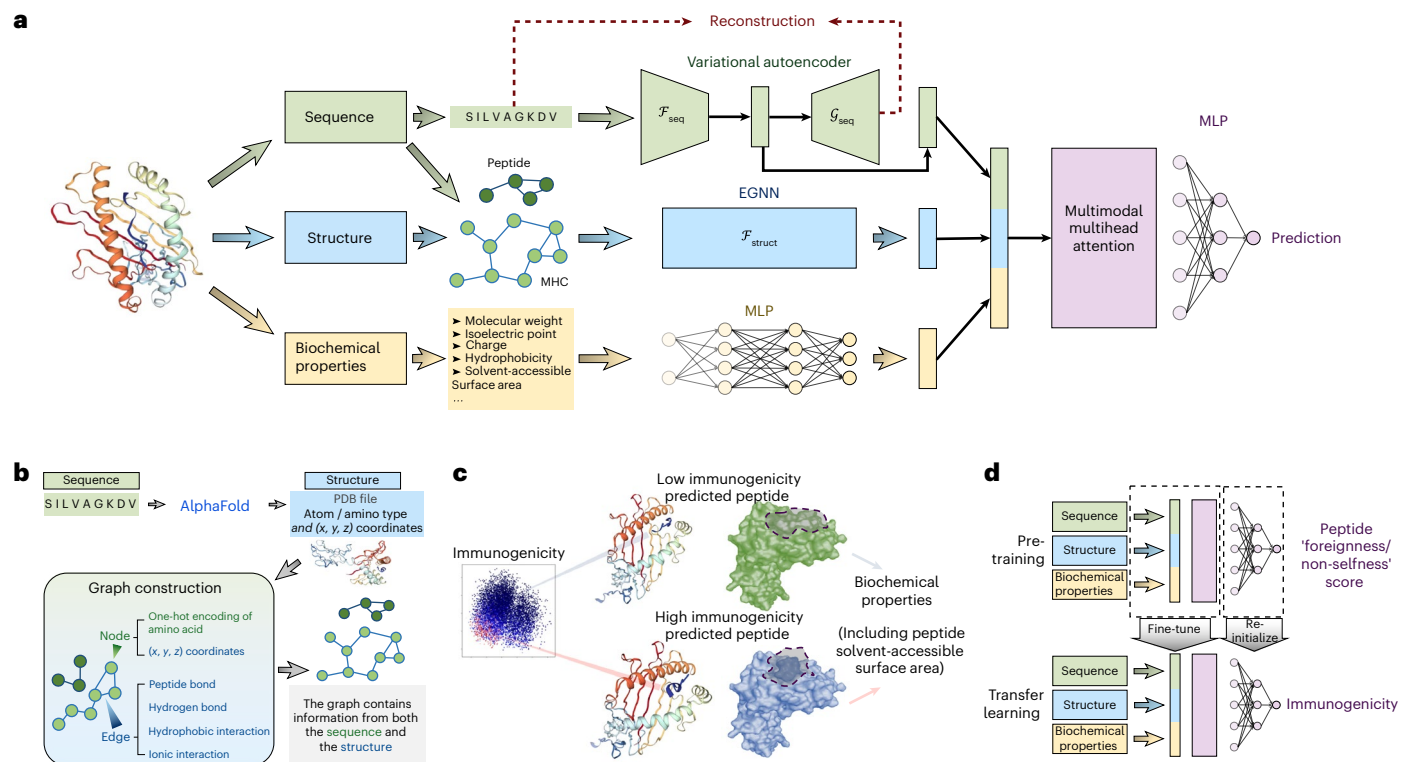


Fig. 1 | Overview of ImmunoStruct architecture and training. a, ImmunoStruct separately encodes three modalities and uses the fused multimodal representations for downstream prediction. **b**, Peptide–M2HC graphs are constructed from AlphaFold2 output. Graph edges encode rich biochemical bond information within the peptide–MHC structure, while the nodes encode

the 3D coordinates and the amino acid sequence. **c**, Biochemical properties are computed from the peptide–MHC structure and peptide sequence. **d**, During transfer learning, the model is pretrained to predict peptide foreignness and further tuned to predict immunogenicity.

the peptide, which is derived and processed from the full protein, and then concatenated to the end of the MHC sequence. We speculated that directly encoding the full peptide–MHC sequence information into a more compact and regularized low-dimensional space would enable learning additional sequence-based patterns and relationships across similar sequences and motifs. By encoding the full peptide–MHC sequences in this compact representation, we obtain a 32-dimensional latent space for each sample (Fig. 1a) and reverse map this to the input dimension to perform sequence reconstruction. This architecture choice is helpful for learning a compact and meaningful latent space that is semantically continuous across samples^{28–31}.

Biochemical property modelling. Given the known biochemical involvement of interactions within the peptide–MHC–TCR synapse, and the importance of surface-area accessibility of the loaded peptide for immunogenicity^{12,17}, we used the structural data for each peptide–MHC sample to calculate the solvent-accessible surface area of the peptide–MHC to include as a component of the biochemical features (Fig. 1c). The biochemical properties also included properties of each peptide, such as molecular weight, isoelectric point, charge and hydrophobicity^{7,32}. The properties are then fed into the network as additional features (Fig. 1a).

Feature combination via multimodal multihead attention. The 32-dimensional sequence embedding and the 64-dimensional structural embedding, along with the biochemical property embedding, are concatenated to form a 104-dimensional representation vector (Fig. 1a). This multimodal representation undergoes further processing with a multihead attention layer, which consists of eight self-attention heads applied directly to the multimodal representation vector. Attention visualization showed learned multimodal attention weights across

the concatenated embedding, indicating information was attended to across each input modality (Supplementary Fig. 1c).

ImmunoStruct training

Transfer learning. ImmunoStruct is trained using transfer learning³³ (Fig. 1d). After weight initialization, it is first pretrained to predict foreignness, a measure that quantifies the similarity between a peptide and sequences of known pathogens, foreign to humans, which can be associated with immunogenicity^{7,34}. The score is computed using a multistate thermodynamic model as previously described³⁵. After that, it is fine-tuned to predict immunogenicity, which is a binary experimental readout obtained from a T cell recognition or activation assay that specifically detects recognition or activation of CD8⁺ T cells to a given peptide–MHC. All modules are warm-started from the pretrained weights, with the exception of the final MLP, which is re-initialized to accommodate the change in task (Fig. 1d). The transfer learning process leverages patterns between foreignness and epitope recognition^{34,36} to improve the performance on the main immunogenicity task that is different but related to the pretraining task. For both infectious disease and cancer models, graph data were generated from predicted AlphaFold2 structures augmented by sequences as the node features (Fig. 1b), while biochemical property data were derived from sequence and structural surface features (Fig. 1c). Details are further described in Methods.

Overview and cancer–wild-type peptide–MHC contrastive learning. Immune responses to cancer neopeptides differ fundamentally from those to infectious epitopes, motivating a cancer-specific contrastive learning strategy in ImmunoStruct. The framework is grounded in the immunological principle of central and peripheral tolerance: during thymic development, T cells that strongly recognize self-derived

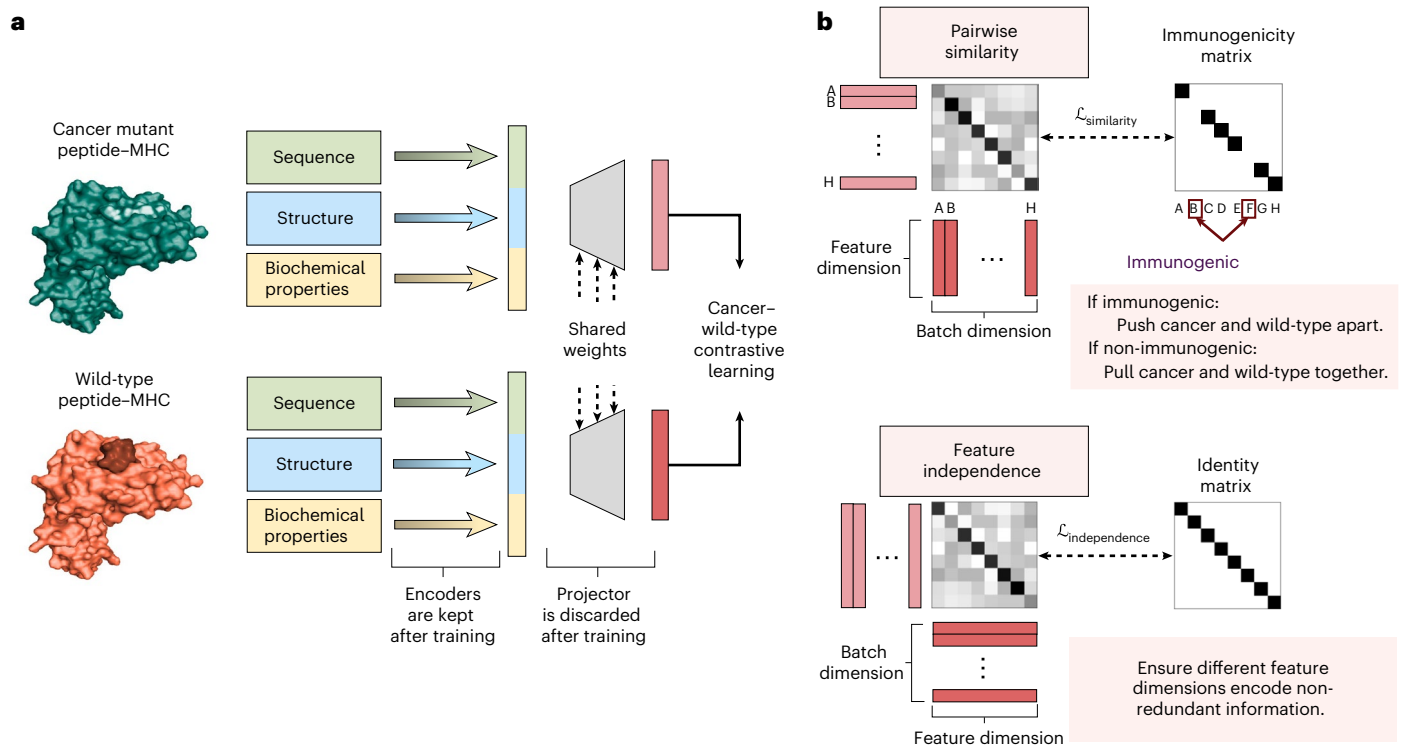


Fig. 2 | Cancer–wild-type contrastive learning leverages the wild-type counterpart to better organize the latent space. a, ImmunoStruct extracts multimodal latent representations by integrating sequence, structure and biochemical features. Cancer–wild-type contrastive learning optimizes these representations by contrasting cancer mutants with their wild-type

counterparts. **b**, Cancer–wild-type contrastive learning consists of two complementary components: the pairwise similarity component pulls or pushes latent representations depending on the immunogenicity of the peptide–MHC, and the feature-independence component encourages different dimensions of the representations to encode diverse and distinct information.

peptide–MHCs are deleted or anergized^{12,37}. As a result, wild-type peptide–MHCs are generally immunologically silent. Mutations, on the other hand, can modify TCR-facing residues. Mutant peptide–MHCs not encountered during thymic selection may bypass tolerance and, in turn, elicit T cell responses. In summary, tolerance enforces similarity between non-immunogenic mutants and the wild-type, while immunogenic mutants must diverge from the wild-type. This reflects previous evidence that neoantigen ‘non-selfness’, quantified by dissimilarity to the self-proteome or by mutant–wild-type binding gain, correlates with T cell recognition^{34,38}.

Building on this biology, we introduce a cancer–wild-type contrastive learning objective tailored to mutant–wild-type pairs. Specifically, the model learns to embed mutant peptides close to their wild-type counterparts when the pair is non-immunogenic, but to separate them when the mutant is immunogenic (Fig. 2a). Besides this immunogenicity-conditioned push–pull objective, the contrastive learning objective has a second feature-independence component that promotes independence across feature dimensions and prevents feature collapse (Fig. 2b). Details are further described in Methods.

Improving immunogenicity prediction on infectious disease epitopes

We first evaluated ImmunoStruct for its ability to predict peptide–MHC immunogenicity on the IEDB dataset of infectious disease epitopes, and compared it with existing methods including Prime-2.0²², NetMHCpan¹³, MHCnuggets¹⁴, MHCflurry¹⁵, BigMHC¹⁶ and DeepNeo²³. Performances are reported on area under the receiver–operating characteristic (AUROC) curve, area under the precision–recall curve (AUPRC) and mean PPVn. AUROC and AUPRC are composite metrics that reflect the threshold-agnostic aggregated classification performance on {sensitivity, specificity} and {precision, recall}, respectively. Mean PPVn is

a widely used and clinically important metric^{13–16} that measures the precision of several top-ranked peptide–MHC candidates predicted in a setting similar to epitope ranking for clinical trial implementation.

For all metrics, ImmunoStruct established a state of the art, with an AUROC of 0.882 ± 0.005 , an AUPRC of 0.696 ± 0.020 and a mean PPVn of 0.514 ± 0.020 , as shown in Fig. 3a. Notably, on the key measure of AUPRC, ImmunoStruct (0.696) largely outperformed PRIME2.1 (0.212), NetMHCpan (0.214), MHCnuggets (0.246), MHCflurry (0.260), DeepNeo (0.269), BigMHC-EL (0.319) and BigMHC-IM (0.462). The results can be found in Fig. 3a and Supplementary Table 3.

To further assess the robustness of ImmunoStruct across allele frequencies, we performed an additional allele-aware evaluation. In this setting, samples corresponding to the ten lowest-frequency alleles were completely excluded from training and used only for testing (Supplementary Table 4). Despite being trained solely on common alleles, ImmunoStruct maintained strong predictive performance on these held-out rare alleles, indicating that the model generalizes effectively beyond the alleles represented during training.

To verify that all input modalities are beneficial and synergistically improve model performance, we conducted an ablation experiment (Fig. 3b and Supplementary Table 1). The ImmunoStruct model that takes in all input modalities outperformed all of its ablated counterparts and showed a synergistic pattern on including all components and modalities. Specifically, this showed that each modality was necessary for optimal performance when combined with the transfer learning approach. The exclusion of structure, biochemical properties or sequence generally resulted in a clear drop in performance across test splits. This analysis also demonstrated that transfer learning is generally beneficial compared with training from scratch.

The same phenomenon can be observed qualitatively as well. By visualizing the latent space embeddings using PHATE³⁹, we observed

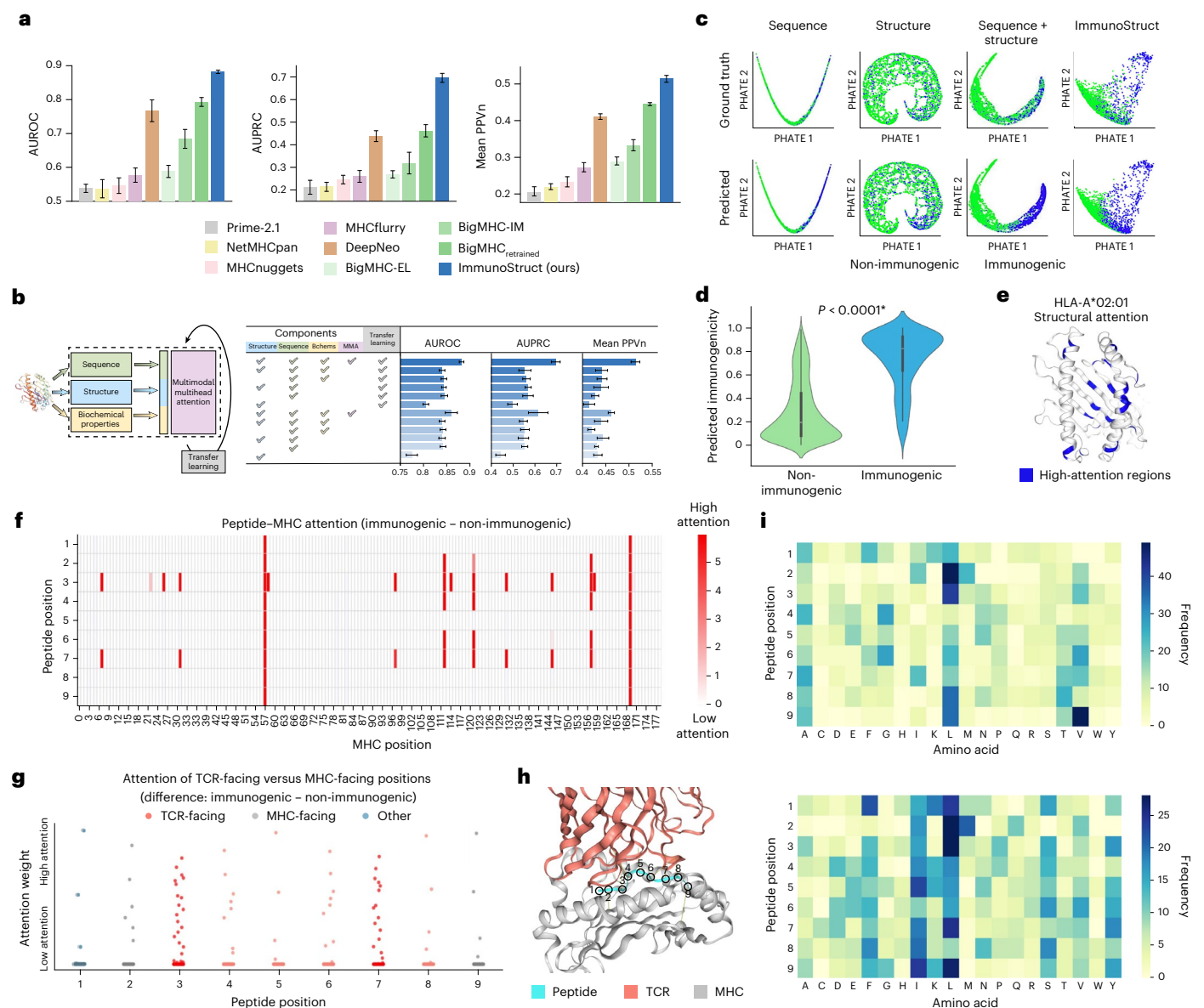


Fig. 3 | ImmunoStruct improves immunogenicity prediction on the IEDB dataset of infectious disease.

a, The performance of ImmunoStruct compared against existing methods, evaluated on the IEDB test set with $n = 3,500$ peptide–MHC samples. Bars indicate mean values and error bars indicate sample s.d. (for AUROC, AUPRC) or s.e.m. (for mean PPVn) from five independent experiments. For trainable models, results are reported on the test set after the models are trained on the non-overlapping training data. For pretrained models with public API but without open-source code, evaluations are performed on the five test sets. **b**, Ablation studies on the test set with $n = 3,500$ peptide–MHC samples suggest that transfer learning is helpful and that combining all components with multimodal multihead attention leads to the best performance. Bars indicate mean values and error bars indicate sample s.d. (for AUROC, AUPRC) or s.e.m. (for mean PPVn) from five independent experiments. **c**, Embedding visualizations between ground-truth (top) and predicted (bottom) immunogenicity scores suggests that the ImmunoStruct

model learns a structured latent space that can well distinguish immunogenic and non-immunogenic samples by fusing multimodal information (sequence, structure and biochemical properties). **d**, Immunogenicity scores predicted by ImmunoStruct over all $n = 3,500$ peptide–MHC samples in the test set show a clear separation between immunogenic and non-immunogenic peptide–MHCs using a two-sided t -test. **e**, The structural attention visualizations shows the high-attention locations of the MHC protein structure. **f**, Learned immunogenicity-enhanced attention between peptide and MHC positions. **g**, Averaged attention across peptide positions. The TCR-facing positions 3 and 7 show high attention. **h**, The structural visualization shows that peptide positions 2 and 9 are MHC-facing and peptide positions 3 to 8 are TCR facing. More views are available in Supplementary Fig. 3. **i**, Heatmap of amino acid frequencies in high-immunogenicity (top) versus low-immunogenicity (bottom) peptide–MHCs. **e–i** are generated on allele HLA-A*02:01.

that the latent space of ImmunoStruct with all three modalities exhibited the clearest organization when comparing the true versus predicted labels (Fig. 3c). These visualizations indeed show an increasingly organized embedding space of peptide–MHC immunogenicity, coloured by true immunogenicity (top row) versus ImmunoStruct predictions (bottom row). This embedding organization becomes most pronounced for the ImmunoStruct model.

We next leveraged the ImmunoStruct EGNN attention layer to assess which positions on both the peptide and MHC were important for prediction. For easy comparison of protein index positions, we performed this analysis on a single allele (HLA-A*02:01). We first show strong separation in ImmunoStruct prediction scores between non-immunogenic and immunogenic samples (Fig. 3d). Next, we extracted the learned structural attention weights and overlaid them

onto the MHC structure, showing the high-attention MHC enrichment throughout the $\alpha 1$ and $\alpha 2$ peptide-binding cleft domains, and the N-terminal extracellular domain (Fig. 3e).

When assessing peptide level structural attention weights (Fig. 3f,g), we observed enrichment in learned attention values at TCR-facing positions—particularly positions 3, 4, 6 and 7 of the peptide—as opposed to MHC-facing positions (for example, 2 and 9), which would be more related to peptide–MHC binding rather than immunogenicity^{40–44} (Fig. 3h). On assessing these positions more closely, we found that immunogenic samples display less diversity at these TCR-facing positions (Fig. 3i). Visualization of these positions on a known example peptide–MHC–TCR structure also show protrusion of these peptide positions towards the TCR, which aligns with the critical role of TCR interactions in the immunogenic response and may suggest increased sequence–structural focus towards these positions in immunogenic samples (Fig. 3h). To further validate the top attention positions against known TCR-contacting residues, we selected the HTLV-1 Tax peptide–MHC–TCR complex (Protein Data Bank (PDB) 1BD2) as our reference system. The Tax peptide is a highly immunogenic antigen derived from an oncogenic virus and has been functionally characterized in complex with the A6 TCR⁴⁵. In line with the learned attention signals, inspection of the Tax–A6–HLA–A2 complex revealed that the central portion of the peptide projects upward towards the TCR, producing a characteristic bulge across positions between P2 and P9 (Supplementary Fig. 2a). Quantitative spatial distance analysis of peptide–TCR geometry supported this observation: the shortest C α –C α distances were concentrated at the central residues, while the termini remained farther away (Supplementary Fig. 2b). These results provide a structural ground-truth mapping for our learned attention signals with known contacting spatial distances. This also suggests that the model learned that features of a bulged peptide backbone probably incorporate key components determining whether or not a peptide–MHC is likely to be immunogenic.

We also observed differences in amino acid frequencies between immunogenic and non-immunogenic peptides (Fig. 3i), as well as a higher enrichment of alanine at position 7 and preference of leucine at position 3. The TCR-facing and MHC-facing positions are illustrated in Fig. 3h, with more views available in Supplementary Fig. 3. Together, these results indicate ImmunoStruct could be leveraged to pinpoint certain peptide–MHC positions that are important for immunogenicity prediction.

Improving immunogenicity prediction on human cancer neoepitopes

The clinical potential of therapeutic cancer neoepitope-based vaccines has been increasing in recent years^{1–3}, so we next sought to explore the prediction performance of ImmunoStruct in the setting of cancer neoepitope immunogenicity. This task may improve prioritization of the most immunogenic peptides for epitope-based vaccine design, which is restricted to a limited number of peptide candidates. Missense mutations to a protein can lead to presentation of a peptide that only differs by one amino acid from its wild-type form, which may make distinguishing immunogenic neoepitopes challenging^{12,46}.

On extending ImmunoStruct to this data setting, we observed superior performance compared with existing methods (Fig. 4) despite a relatively smaller training set than some other competitors. ImmunoStruct was able to outperform existing methods on AUROC, AUPRC and mean PPVn (Fig. 4a and Supplementary Table 3). We also showed through an ablation study that the cancer–wild-type contrastive learning framework and transfer learning both additively improved ImmunoStruct performance (Fig. 4b). In addition, we conducted an ablation study on cancer–wild-type contrastive loss components to demonstrate that the two components are complementary to the overall success (Supplementary Table 2).

Given the structural subtleties in cancer neoantigen peptide–MHC immunogenicity particularly for oncogene mutant isoforms, we next sought to identify and explore the ability of ImmunoStruct to properly capture these differences and result in appropriate predictions. To that end, we first ran ImmunoStruct on the KRAS–G12V and KRAS–G12D peptide–MHC (HLA-A*11:01) mutants as a test case analysis, resulting in increased immunogenicity predictions for the G12V mutant compared with the G12D mutant (0.64 versus 0.48), as visualized in Fig. 4c. Indeed, when experimentally measured within the clinical neoantigen vaccine context (NCT03953235, ref. 47), the KRAS–G12V mutant displayed increased immunogenicity compared with the KRAS–G12D isoform across three separate vaccine cassettes ($P < 0.05$, Fig. 4d). These results indicate the ability of ImmunoStruct to capture subtle peptide–MHC multimodal differences resulting from a single isoform amino acid change from the same protein.

More importantly, we next sought to explore which parts of the peptide–MHC structure contribute the most to the prediction of immunogenicity and to visually compare the differences between the two isoforms. We extracted the learned ImmunoStruct attention weights from the structural multihead attention layers. So far and to our knowledge, there have been no structural graph attention-based deep learning methods from wide-scale peptide–MHC structure data for immunogenic biostructural interpretation. We therefore built an attention visualization module that helps highlight critical regions of the peptide–MHC that influence the predicted immunogenicity, providing insights into structural determinants of epitope recognition. When extracting the learned attention weights, we observed a region of the peptide–MHC groove surface that showed differential sterics between the two isoforms that aligned with the learned attention weight at those positions (Fig. 4e). This suggested that towards the terminal end of the KRAS–G12V peptide presented, a differential peptide–MHC interaction may contribute in part to a higher immunogenic response, as has been observed in the context of other peptide–MHCs⁴⁸. Additional example cases of structural attention from the visualization module are shown in Supplementary Fig. 1.

Experimental validation on SARS-CoV-2 epitopes and clinical cancer survival prediction

Given the importance of further validation for an immunogenicity prediction model, we sought to validate our predictions with ELISpot IFN γ measurements that we generated in vitro for a set of peptides in combination with one of five HLA alleles. ImmunoStruct successfully predicted 15 out of 19 immunogenicities (AUROC = 0.780, AUPRC = 0.941 and accuracy 79.0%) (Fig. 5a). For each allele, one representative plate is shown to visualize IFN γ production for a given peptide–MHC (Fig. 5b), further described in Methods.

To extend the utility of ImmunoStruct to the clinical cancer setting, we validated ImmunoStruct in a cohort of patients with cancer with available peptide–MHC data, comprising a variety of different tumour types and immunotherapeutic treatments⁴⁶. In comparison to tumour-mutational burden, we found ImmunoStruct predictions were able to stratify patient survival more effectively for both overall survival and progression-free survival without any retraining (hazard ratio 0.54 (0.30–0.97) and hazard ratios 0.54 (0.31–0.93), respectively, Fig. 5c,d). Cohort stratification was done by separating high versus low patient groups using the median predicted total count of immunogenic peptide–MHCs across all patients. We also show comparisons with existing methods, indicating improved performance on peptide–MHC-based prediction of overall survival of patients with cancer and progression-free survival (Fig. 5c,d and Supplementary Fig. 4). Together, these results indicate the ability of ImmunoStruct to capture clinically relevant immunogenic peptide–MHCs that associate with survival in the context of cancer immunotherapy.

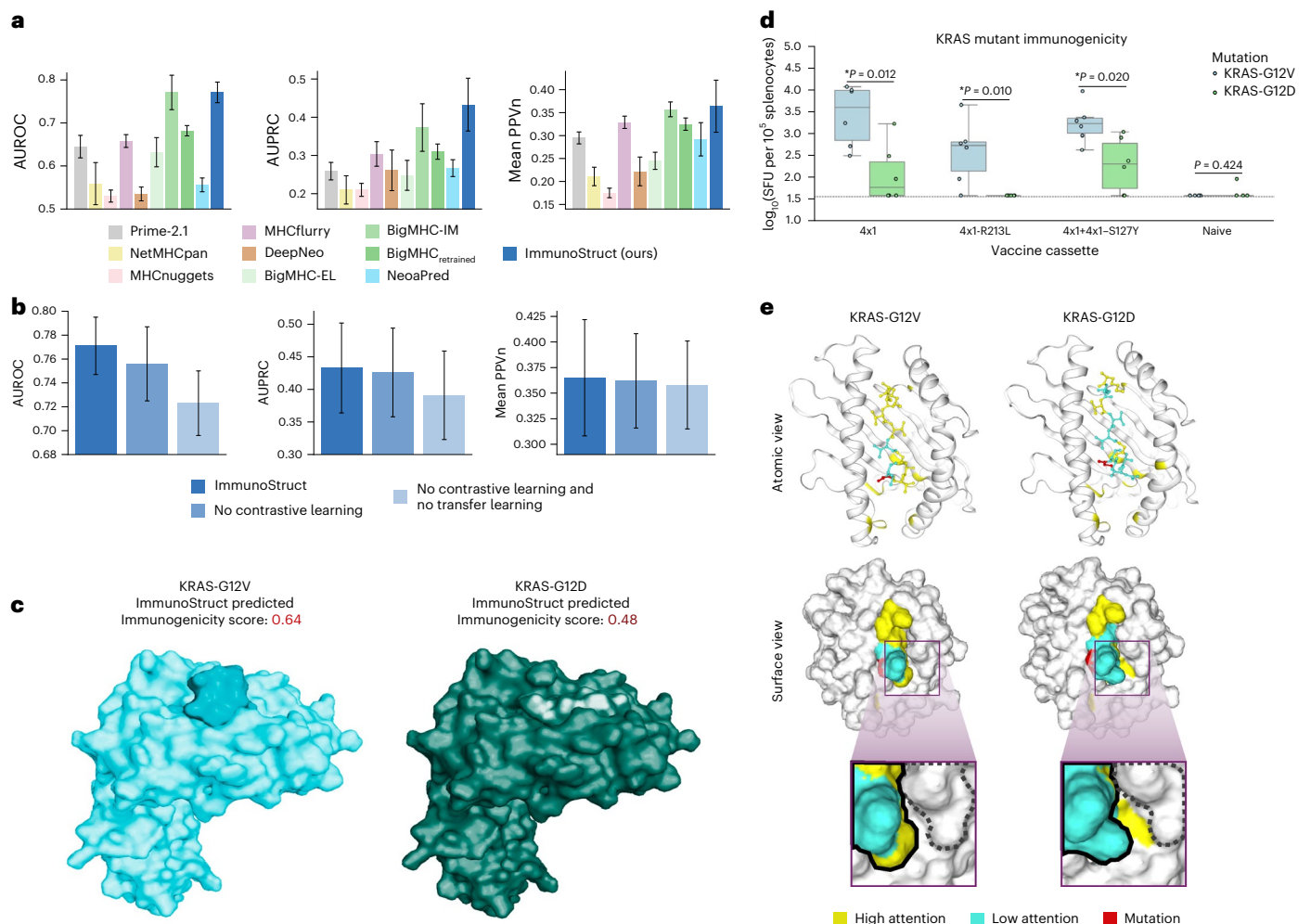


Fig. 4 | ImmunoStruct improves immunogenicity prediction on the CEDAR dataset of cancer neoepitopes. a, The performance of ImmunoStruct compared against existing methods, evaluated on the CEDAR test set with $n = 407$ peptide-MHC samples. Bars indicate mean values and error bars indicate the sample s.d. (for AUROC and AUPRC) or the s.e.m. (for mean PPVn) from five independent experiments. For trainable models, results are reported on the test set after the models are trained on the non-overlapping training data. For pretrained models with public API but without open-source code, evaluations are performed on the five test sets. **b**, Ablation studies on the test set with $n = 407$ peptide-MHC samples suggest that transfer learning and cancer-wild-type contrastive learning are both helpful. Bars indicate mean values and error bars indicate

sample s.d. (for AUROC, AUPRC) or s.e.m. (for mean PPVn) from five independent experiments. **c**, Protein molecular surface visualization of KRAS mutants with ImmunoStruct immunogenicity scores. **d**, Experimental KRAS mutant isoform immunogenicity for three distinct vaccine constructs containing either the G12V or G12D isoform or a naive negative control. Each dot represents an individual sample measuring IFN γ responses or antigen-specific T cell responses to KRAS mutant peptides. Box plots show the median (centre line) and interquartile range (box), and whiskers extend to the minimum and maximum values. Reported P values are from two-sided t -tests. SFU, spot forming units. **e**, KRAS mutant peptide-MHC visualization with ImmunoStruct attention overlay. Solid lines outline the peptide; dotted lines outline nearby MHC protein residues.

Discussion

The ImmunoStruct model is a jointly trained structure-sequence model that uses large-scale multi-allelic structural and biochemical peptide-MHC data for application to the immunogenicity prediction task, across both infectious disease and cancer neoepitopes. ImmunoStruct uses the graph transformer, MLP and VAE neural networks for different modalities, culminating in a multimodal architecture capable of synergistically integrating three distinct data modalities for this complex task. Second, within the graph transformer, the multihead self-attention mechanism provides biostructural insights for immunogenic peptide-MHC features and the ability to visualize regions of interest in a peptide-MHC complex. Third, the contrastive loss approach for cancer neoepitopes improves prediction performance by embedding wild-type and mutant peptide sequences for immunogenic peptide-MHCs far despite differing by only a single amino acid. Last, we also provide a curated multimodal dataset containing structural, sequence and biochemical property information for each labelled peptide-MHC sample.

When assessing the design steps and modalities of ImmunoStruct, each component was biologically inspired, aiming to extract unique information from each peptide-MHC data type. For example, the structure embedding encoded from the graph transformer module allowed for precise encoding of spatial relationships that are relevant for TCR binding to the peptide-MHC surface¹². The peptide-MHC surface characteristics were further accounted for by the solvent-accessible surface-area feature included in the model. Such information may otherwise be overlooked by sequence-only models. The pretraining on peptide foreignness was also inspired by the concept of T cell self-tolerance mechanisms and higher observed immunogenicity to foreign peptides^{7,34,36}, showing an added increase in performance. For structures we generated from AlphaFold, our goal was to determine whether there would be a net change in immunogenicity prediction performance with the added structural information. We learned through our ablation studies that inclusions of all components together with transfer learning, synergistically, rather than additively, improved

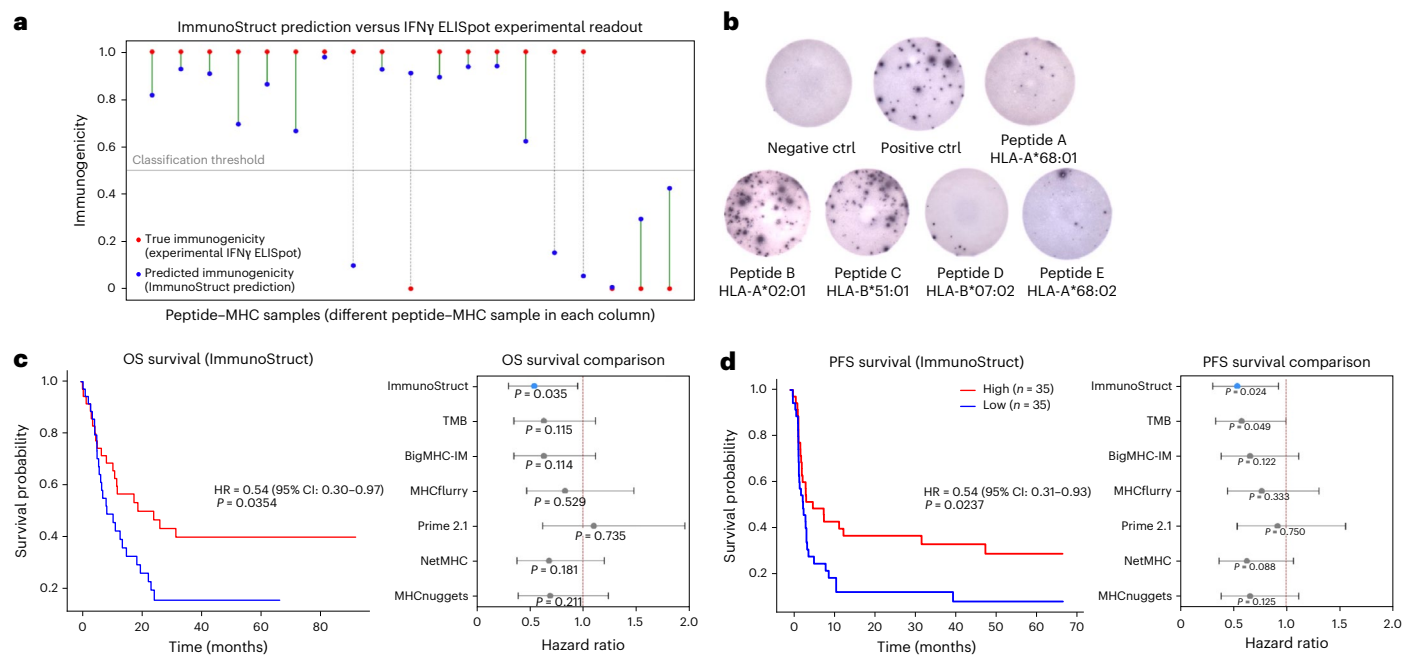


Fig. 5 | Experimental validation of ImmunoStruct predictions. **a**, Lollipop plot where each column represents a unique epitope-MHC pair. Red dots indicate the experimentally measured immunogenicity with IFN γ ELISpot and blue dots indicate the predicted immunogenicity from ImmunoStruct. Vertical lines connect paired values for each sample, with green lines for correct predictions and dashed grey lines for incorrect predictions. **b**, ELISpot plate images with each plate corresponding to a validated allele. **c, d**, Kaplan–Meier survival analyses on $n = 70$ patients with cancer undergoing immunotherapy, stratified by the median count of ImmunoStruct-predicted immunogenic peptide–MHCs into

high-immunogenicity ($n = 35$ patients) and low-immunogenicity ($n = 35$ patients) groups. ImmunoStruct-predicted immunogenicity demonstrates strong association with survival (left), outperforming existing methods (right). **c**, Overall survival (OS) (ImmunoStruct). **d**, Progression-free survival (PFS). In the survival comparisons, centres are hazard ratios and horizontal error bars indicate 95% confidence intervals. The separation between high and low groups is reported in the P value using a two-sided t -test. Survival curves for all methods compared can be found in Supplementary Fig. 4.

performance. This may suggest that there exist nonlinear combinations of learned downstream feature vectors that contain meaningful information for future exploration of immunogenicity prediction.

When specifically exploring cancer neoepitopes, we added a conditional contrastive loss to ensure that the mutant peptide will be embedded far from its wild-type counterpart only if the mutation results in immunogenicity. This helped force the model to learn how to differentiate each pair and capture subtle differences in peptide–MHC variants. We applied this to a case study for the known and highly prevalent cancer oncogene *KRAS*, which is a clinical target across several solid tumour types⁴⁷. Specifically, we saw that when leveraging experimentally validated *KRAS*-G12V versus *KRAS*-G12D with in vivo samples, we were able to leverage our attention-based visualization module to highlight subtle peptide–MHC spatial proximity that differed within the peptide-binding groove between each complex. The attention difference captured was at the C-terminal end of the peptide rather than at the mutation itself, suggesting a conformational peptide–MHC effect rather than the mutational effect at the sequence level itself. In our *KRAS* analysis, it is also interesting to note that HLA-A*11:01-restricted G12V has demonstrated robust T cell recognition with structural and functional validation, whereas G12D/A*11:01 has progressed via engineered high-affinity TCRs and early clinical translation^{47,49,50}. Across A*11:01-restricted reports, we note a trend toward greater observed T cell immunogenicity for *KRAS*-G12V than G12D, with higher frequency and avidity of natural responses and broader immune recognition. However, direct head-to-head clinical comparisons remain limited and context-dependent. For example, in unresectable pancreatic ductal adenocarcinoma, recent work reported shorter overall survival for *KRAS*-G12D/G12R than for G12V, suggesting comparatively better survival associated with G12V within this clinical context^{47,49–51}. Experimental observations in line with our observed structural attention findings

surfaced from ImmunoStruct may encourage interesting avenues of machine-guided future exploration in the domain of cancer neoantigen vaccine design as well as TCR design. In addition, given these subtle conformational differences may exist across different samples and MHC alleles, a wide-scale generation and integration of molecular dynamics peptide–MHC data may also provide an interesting future direction of our work.

Despite these contributions and interesting avenues of future exploration, the study has limitations. First, as a general limitation in the field of immunogenicity prediction, the peptide–MHC data used for training relies on experimental labels, which are inherently variable to some degree owing to differences in assay protocols, cell lines and experimental designs. Second, while TCR information would allow more information for prediction, the astronomical diversity of the human TCR repertoire across peptide–MHCs limits training data availability. Third, ImmunoStruct was trained on only the 27 most common HLA allele protein sequences⁵², and therefore further analysis would be required to explore its application to rare alleles with low training data. In addition, as with all public peptide–MHC data, our dataset is skewed toward a subset of common class I alleles (for example, HLA-A*02:01, HLA-A*11:01) and more commonly evaluated pathogen peptide sequences, which can bias both representation and performance of the model. To address this, we provide per-allele breakdowns and evaluate generalization to rare or held-out alleles that show a reasonable retention in performance. Nevertheless, we acknowledge that residual study-specific and assay-specific biases probably persist from data coverage for a given allele or pathogen class. Finally, the application to clinical neoepitope survival consisted of a small sample size and was not homogenized with regard to a single immunotherapy type. However, this also highlights the need for additional publicly available clinical-genomic immunotherapy datasets for validation of tools within this field.

Together, ImmunoStruct provides a multimodal structural, sequential and biochemical approach that is meaningfully able to jointly model protein structure and sequence for the challenging task of immunogenicity prediction. With the advent of mRNA cancer neo-antigen vaccines, as well as immunogenicity bio-therapeutic toxicity challenges, this work has applications to not only epitope-based vaccines, but also protein-based therapeutic discovery. More generally, it may also provide a framework for protein-complex property prediction and protein design goals that may benefit from integrating biochemical and predicted structural data.

Methods

ImmunoStruct

Structure–sequence modelling with multimodal graph transformer. To create the peptide–MHC structure dataset that also contains sequence features, we first use AlphaFold2²¹ to generate PDB files containing the folded protein structures from the peptide–MHC sequences. For each peptide–MHC sequence, the protein structures are represented as the amino acid residue sequence information and their respective (x, y, z) coordinates in the 3D space. First, we trim the structure to take in consideration only the extracellular domain of the peptide–MHC. We then construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where the nodes $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$ are the one-hot encoding of the amino acid residues concatenated to their (x, y, z) coordinates, whereas the edges $\mathcal{E} = \{e_1, e_2, \dots, e_m\}$ are binary existence indicators of peptide bonds, hydrogen bonds, hydrophobic interactions or ionic interactions between two vertices. Specifically, a node v_i is represented as a 1×23 vector $[AA_1, \dots, AA_{20}, x, y, z]^T$ where $AA_{[0:20]}$ one-hot encodes the amino acid residue. An edge e_{ij} is defined as 1 if there exists any aforementioned bond or interaction between v_i and v_j , and as 0 otherwise.

When modelling the peptide–MHC structures, ImmunoStruct leverages a graph transformer module with $E(n)$ -equivariant layers (EGNN)²⁴ and multihead self-attention (later described in section ‘Multimodal data integration strategy’) across all node–sequence and spatial information of the peptide–MHC structure. The EGNN layers enable powerful processing of data represented as graphs, where nodes correspond to amino acid residues and edges represent the interactions among residues. EGNN layers further amend the graph neural network by ensuring equivariance to 3D coordinates. For this task these equivariant layers are suitable because they are equivariant to rotations, translations, reflections and permutations so the arbitrary positions and orientations from the AlphaFold 3D coordinates would otherwise be properly handled.

The propagation rules from layer l to $l+1$ is described in equation (1). h is the node embedding, x is the coordinate embedding, a is the edge attributes and the ϕ s are learnable operations. Following the official implementation²⁴, C is set to be $\frac{1}{M-1}$, where M is 3 for the 3D coordinates.

$$\begin{aligned} m_{ij} &= \phi_e(h_i^l, h_j^l, \|x_i^l - x_j^l\|^2, a_{ij}) \\ x_i^{l+1} &= x_i^l + C \sum_{j \neq i} (x_i^l - x_j^l) \phi_x(m_{ij}) \\ m_i &= \sum_{j \neq i} m_{ij} \\ h_i^{l+1} &= \phi_h(h_i^l, m_i) \end{aligned} \quad (1)$$

In our case, h_i^l is the embedding of residue v_i at layer l , x_i^l are the coordinate embeddings from spatial coordinates (x, y, z) and a_{ij} are the edge attributes as previously described.

The combination of multihead self-attention and graph processing layers ultimately enable the model to capture the spatial and pairwise relationships across the peptide–MHC, which are cooperatively recognized by a complementary TCR structure within the immunogenic synapse.

Sequence modelling with a VAE. The peptide–MHC sequences are encoded via a VAE²⁸. The VAE learns a continuous and structured latent space by introducing a probabilistic framework. It uses an encoder g_ϕ and a decoder f_θ to encode the input x to a latent vector z and reconstruct it back to $\hat{x}$, which approximates the input. In a standard autoencoder, the process is described in equation (2) and the model is optimized by minimizing the L2-distance between the real and reconstructed input $\|x - \hat{x}\|$.

$$\begin{aligned} z &= g_\phi(x) \\ \hat{x} &= f_\theta(z) \end{aligned} \quad (2)$$

VAE introduces a probabilistic framework by separately modelling the mean and variance of the data, as described in equation (3), where $\odot$ stands for element-wise product. Besides minimizing the L2-distance $\|x - \hat{x}\|$, it also minimizes the Kullback–Leibler (KL) divergence between the latent embedding z and a standard normal distribution $\mathcal{N}(0, \mathbf{I})$. This allows the VAE to learn a semantically continuous latent space, since latent embedding z are reconstructed to approximate the same input despite small perturbations $\sigma \odot \epsilon$. For more details on the optimization, please refer to section ‘ImmunoStruct optimization objective’.

$$\begin{aligned} \mu, \sigma &= g_\phi(x) \\ z &= \mu + \sigma \odot \epsilon, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}) \\ \hat{x} &= f_\theta(z) \end{aligned} \quad (3)$$

In our implementation, the VAE module projects the higher-dimensional sequence data into a lower 32-dimensional latent space. Because of the semantic continuity property, similar peptide–MHC sequences are embedded closely. This is beneficial to the immunogenicity prediction task as sequence-based motifs are known to associate with immunogenicity. Last, the semantically continuous latent space is easily amendable to downstream data integration with other data modalities, namely structure and biochemical properties.

Biochemical property modelling with an MLP. An MLP that consists of two linear layers with rectified linear unit activation and a dropout layer is used to extract an embedding vector from the raw biochemical property values. This light-weight module allows the learning of higher-order nonlinear combinations of these properties that serve as helpful features that are biochemically derived for the downstream task.

Transfer learning over peptide foreignness. Before training the model to perform the binary classification task of immunogenicity, our model is first trained to predict foreignness scores as defined in refs. 7,34,36. The foreignness can be seen as a measure of how similar a sequence is to sequences of known pathogens and was shown to be related to immunogenicity.

As defined by^{35,36} foreignness (R) follows a thermodynamic model as:

$$R = \frac{\sum_{e \in E} e^{-k(a-|s,e|)}}{1 + \sum_{e \in E} e^{-k(a-|s,e|)}} \quad (4)$$

We used the same parameters as ref. 35 Where E is the dataset of pathogen sequences, $k = 4.87$, $a = 26$ and $|s, e|$ defines how similar in sequence s is from e using computed alignment scores.

This score enabled us to first train a regressor and then perform the final task. The ablation studies (Fig. 3a) showed that this transfer learning method improved performance.

Multimodal data integration strategy. After encoding of each data-modality in latent space, with $h_{\text{sequence}} \in \mathbb{R}^{32}$, $h_{\text{structure}} \in \mathbb{R}^{64}$, $h_{\text{bchems}} \in \mathbb{R}^8$, the modality-specific embeddings are concatenated into

a single input vector $\mathbf{X} = h_{\text{sequence}} \parallel h_{\text{structure}} \parallel h_{\text{bchems}}$, $\mathbf{X} \in \mathbb{R}^{104}$. This concatenation aligns the features from heterogeneous sources into a single common embedding vector, ensuring that the downstream layers can jointly input the information from each modality. This vector is then processed using a multihead self-attention mechanism²⁶ directly on the latent vector.

The standard scaled-dot product attention is performed as follows. We first compute three vectors, namely query $\mathbf{Q}$, key $\mathbf{K}$ and value $\mathbf{V}$ from the input vector $\mathbf{X}$ by matrix multiplications $\mathbf{Q} = W^Q \mathbf{X}$, $\mathbf{K} = W^K \mathbf{X}$, $\mathbf{V} = W^V \mathbf{X}$, where the W s are separate learnable matrices. The attention is given by equation (5), where d_k is the dimension of the key vector.

$$\text{attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (5)$$

$$\text{Softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

Multihead attention²⁶ is an extension to the scaled-dot product attention. Instead of performing a single attention, it projects the queries, keys and values h times with different, learned linear projections (equation (6)). Multihead attention captures diverse aspects of relationships within a sequence by allowing each 'head' to focus on different parts of the data simultaneously.

$$\text{multihead attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1 \parallel \text{head}_2 \parallel \dots \parallel \text{head}_h] W^O \quad (6)$$

$$\text{head}_i = \text{attention}(\mathbf{Q}W_i^Q, \mathbf{K}W_i^K, \mathbf{V}W_i^V)$$

In our implementation, we used a multihead attention mechanism that encompasses eight attention heads to dynamically compute attention weights over the combined multimodal representation. This enables the model to selectively focus on the most informative features for a given sample's prediction, rather than trivially treat the different modalities as if they were equivalent. For instance, in cases where structural nuances have a critical role such as when differentiating subtle peptide–MHC conformations that the peptide–MHC sequences alone would not inherently capture, higher attention weights can be allocated to the structural representations. Furthermore, each attention head can capture distinct cross-modal relationships. This mechanism also inherently provides a form of regularization by allowing the network to down-weight less informative or noisy features from any given modality. This is beneficial in contexts such as ours, where peptide–MHC labels and data quality can vary depending on its source and experimental protocols.

Visualization of structural peptide–MHC attention. A single self-attention layer (as described in equation (5)) was applied directly on the peptide–MHC graphs during training of the EGNN module to capture sequence–structural relationships directly from the peptide–MHC graph. After training, attention weights were extracted in the form of the attention matrix for each peptide–MHC (peptide–MHC) complex. We then isolated the submatrix corresponding to interactions between the peptide and MHC node features, which also contained the sequence information. As the raw attention values are typically very small, we applied a logit transformation across values to enhance subtle differences learned across the peptide. These were then averaged across samples and compared between immunogenic and non-immunogenic complexes across positions of the peptide. For visualization overlays of the attention values, MHC-facing positions were determined on the basis of known literature within the field of peptide–MHC–TCR structural modelling that delineate MHC-facing and TCR-facing positions from ground-truth experimental crystal structure data^{12,41–44}.

ImmunoStruct optimization objective. During training, ImmunoStruct is simultaneously optimized by a prediction loss $\mathcal{L}_{\text{pred}}$, a protein sequence reconstruction loss $\mathcal{L}_{\text{recon}}$, and a normal distribution regularization on the latent space $\mathcal{L}_{\text{normal}}$. The loss function is given by

equation (7), where the loss components are weighted by their respective coefficients $\lambda_{\text{pred}}, \lambda_{\text{recon}}, \lambda_{\text{normal}}$.

$$\mathcal{L}_{\text{ImmunoStruct}} = \lambda_{\text{pred}} \mathcal{L}_{\text{pred}} + \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}} \quad (7)$$

The prediction loss $\mathcal{L}_{\text{pred}}$ guides the model to predict the target accurately, while the reconstruction loss $\mathcal{L}_{\text{recon}}$ ensures that the learned representations carry sufficient information to faithfully reconstruct the input^{30,31,53–55}. In addition, the normal distribution regularization $\mathcal{L}_{\text{normal}}$ encourages the latent space of the VAE to approximate a standard normal distribution with zero mean and unit variance²⁸.

The loss components are shown in equations (8)–(11). $\hat{Y}$ and Y are, respectively, the set of predicted and ground-truth values to be predicted (foreignness or immunogenicity). $\hat{X}$ and X are, respectively, the set of predicted and ground-truth protein sequences. $\hat{Z}$ is the set of latent embedding vectors at the latent space of the VAE, while Z follows the normal distribution $\mathcal{N}(0, \mathbf{I})$. N is the batch size and $|\hat{X}| = |X| = |\hat{Y}| = |Y| = |\hat{Z}| = |Z| = N$.

$$\mathcal{L}_{\text{pred}}^{\text{bin}}(\hat{Y}, Y) = \text{BCE}(\hat{Y}, Y) = \frac{1}{N} \sum_{(\hat{y}, y) \in (\hat{Y}, Y)} -(w_{\text{pos}} y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})) \quad (8)$$

$$\mathcal{L}_{\text{pred}}^{\text{cont}}(\hat{Y}, Y) = \text{m.s.e.}(\hat{Y}, Y) = \frac{1}{N} \sum_{(\hat{y}, y) \in (\hat{Y}, Y)} (\hat{y} - y)^2 \quad (9)$$

$$\mathcal{L}_{\text{recon}}(\hat{X}, X) = \text{m.s.e.}(\hat{X}, X) = \frac{1}{N} \sum_{(\hat{x}, x) \in (\hat{X}, X)} (\hat{x} - x)^2 \quad (10)$$

$$\mathcal{L}_{\text{normal}}(\hat{Z}) = D_{\text{KL}}(\hat{Z} \parallel Z) = \frac{1}{N} \sum_{(\hat{z}, z) \in (\hat{Z}, Z)} \left(\hat{z} \log \frac{\hat{z}}{z} \right) = \frac{1}{N} \sum_{\mathcal{N}(\mu, \epsilon^2) \sim \hat{Z}} (\mu^2 + \epsilon^2 - \epsilon - 1) \quad (11)$$

It should be noted that the prediction loss comes in two alternative forms: When the prediction target is binary (that is, for immunogenicity), it uses the binary cross entropy loss (equation (8)); When the prediction target is continuous (that is, for foreignness), it uses the mean squared error loss (m.s.e.) (equation (9)). The binary cross entropy (BCE) loss is implemented using BCEWithLogitsLoss in PyTorch⁵⁶, where w_{pos} is the positive class weighting coefficient. Following the official recommendation from PyTorch, we set w_{pos} as the ratio of negative to positive sample counts.

For numerical stability, the normal distribution regularization is implemented using the reparameterization trick²⁸, where two separate linear layers in the neural network are respectively used to produce the mean μ and log variance ϵ of the latent representations. Then, the Kullback–Leibler divergence can be computed using summations, subtractions and exponents, without involving divisions (equation (11)).

Cancer–wild-type contrastive learning. When training the cancer ImmunoStruct model, multimodal inputs derived from both the mutant and wild-type peptide–MHCs are leveraged. In practice, a light-weight projector network first projects the multimodal embeddings to a 128-dimensional latent space where cancer–wild-type contrastive learning is performed. The projector, known to be critical for contrastive learning^{57,58}, is only used to indirectly influence the upstream multimodal embeddings and is discarded after training is complete.

As described in equation (12), The optimization objective is a cancer–wild-type contrastive loss $\mathcal{L}_{\text{CW}}$ (equation (13)) scaled by coefficient λ_{CW} and added to the original loss function $\mathcal{L}_{\text{ImmunoStruct}}$. The cancer–wild-type contrastive loss $\mathcal{L}_{\text{CW}}$ consists of two main components, specifically an immunogenicity-aware pairwise similarity loss $\mathcal{L}_{\text{similarity}}$ (equation (14)) and a feature-independence loss $\mathcal{L}_{\text{independence}}$ (equation (15)).

The immunogenicity-aware pairwise similarity loss $\mathcal{L}_{\text{similarity}}$ organizes the latent embeddings across all cancer–wild-type pairs, pulling together non-immunogenic pairs and pushing apart immunogenic pairs. This is to enforce large embedding based distances for mutations that are experimentally observed to result in truly immunogenic neoantigens. The feature-independence loss $\mathcal{L}_{\text{independence}}$ encourages different feature dimensions to encode mutually non-redundant information and therefore mitigates dimensional collapse. The cancer embedding Z_C and the corresponding wild-type embedding Z_W share the dimension $N \times D$, where N is the batch size and D is the feature dimensionality. The immunogenicity vector $\mathbf{1}^C$ is a binary vector of size N , where its element takes a value of 1 if the corresponding peptide–MHC is immunogenic and 0 otherwise.

While the feature-independence loss is inspired by a previous study⁵⁸, the immunogenicity-aware pairwise similarity loss is carefully designed to take the advantage of the unique information from the cancer–wild-type pairs, and is our original contribution.

$$\mathcal{L}_{\text{ImmunoStruct}}^{\text{CW}} = \mathcal{L}_{\text{ImmunoStruct}} + \lambda_{\text{CW}} \cdot \mathcal{L}_{\text{CW}} \quad (12)$$

$$\mathcal{L}_{\text{CW}} = \mathcal{L}_{\text{similarity}} + \mathcal{L}_{\text{independence}} \quad (13)$$

$$\mathcal{L}_{\text{similarity}}(Z_C, Z_W) = \frac{1}{N} \left(\sum_i ((Z_C Z_W^T)_{ii} - \mathbf{1}_i^C)^2 + \lambda_{\text{off-diag}} \sum_i \sum_{j \neq i} ((Z_C Z_W^T)_{ij})^2 \right) \quad (14)$$

$$\mathcal{L}_{\text{independence}}(Z_C, Z_W) = \frac{1}{D} \left(\sum_i ((Z_C^T Z_W)_{ii} - 1)^2 + \lambda_{\text{off-diag}} \sum_i \sum_{j \neq i} ((Z_C^T Z_W)_{ij})^2 \right) \quad (15)$$

Hyperparameters and training details

All experiments were performed on a 40-GB NVIDIA A100 graphical processing unit (GPU), and repeated under five random seeds. For all experiments, we used a batch size N of 128, a learning rate of 1×10^{-3} for pretraining and 1×10^{-4} for fine-tuning. The models were optimized with the AdamW⁵⁹ optimizer with a weight decay of 1×10^{-6} . During pretraining, the learning rate was kept constant for 40 epochs. During fine-tuning, the learning rate was warmed up linearly for 10 epochs, followed by a cosine decay for 30 epochs⁶⁰. For experiments using transfer learning both pretraining and fine-tuning are performed. For experiments marked as ‘no transfer learning’, the pretraining stage is skipped and the model was trained from scratch with fine-tuning.

The coefficients in the loss function (equation (7)) were empirically set to the following values: $\lambda_{\text{pred}} = 2.0$, $\lambda_{\text{recon}} = 0.5$ and $\lambda_{\text{normal}} = 0.5$. In cancer–wild-type contrastive learning (equation (12)), the coefficient was empirically set to $\lambda_{\text{CW}} = 0.01$.

Multimodal dataset for immunogenicity prediction

Owing to the scarcity of publicly available peptide–MHC structural data, we implemented a high-throughput AlphaFold2 pipeline to generate a large peptide–MHC structural dataset. This dataset was ultimately composed of approximately 26,049 entries with paired experimental immunogenicity labels.

Peptide–MHC sequence data. To curate a dataset that would provide sufficient epitope sequence training data across alleles, we downloaded data from the IEDB¹⁹. Peptide sequences were filtered to several criteria: (1) human sequences, (2) T cell experimental assay data (ELISpot, Tetramer staining) and MHC class I restriction. To avoid training across rare human alleles with small sample sizes and often only non-immunogenic cases, we curated a dataset for the top 27 most common alleles in the human population (~97% human population coverage) as previously described^{52,61}. The cancer peptide–MHC sequence data were obtained as previously described and filtered to the same

allele set^{19,46,52,61}. In total, this provided a dataset of 26,049 peptide–MHC pairs with known positive or negative experimental epitope labels. Additional dataset statistics for both the IEDB and the CEDAR datasets are shown in Supplementary Table 5, as well as Supplementary Fig. 5.

Peptide–MHC AlphaFold2 structure data. The AlphaFold2 pipeline was implemented as proposed previously in ref. 21. To run our high-throughput dataset of peptide–MHC sequence combinations, we made use of ColabFold⁶², a wrapper on top of AlphaFold2 that enables highly customizable and easily useable protein folding operations. With ColabFold, we could generate a protein folding from peptide–MHC pairs with only a few lines of code and a shared set of configurations. This enabled us to distribute our protein generation across multiple hosts, drastically reducing the amount of time taken from several weeks to a couple of days. After the peptide–MHC structures were generated, we applied post processing to the structures using the graphein library to convert them to a deep learning model readable graph format⁶³. These processed graph objects contain peptide bonds, hydrogen bonds, ionic interactions and hydrophobic interactions as edges, as well as amino acid node features and their 3D positional information as nodes.

During structure generation with AlphaFold2, crystal templates were not provided as inputs. Only a small fraction of peptide–MHC complexes have experimentally determined crystal structures (less than 1% in our dataset), so template-based prediction was neither necessary nor broadly applicable. To ensure consistency across the dataset, we therefore used template-free AlphaFold2 predictions. Although such predictions may include inherent structural uncertainties, the multimodal design of ImmunoStruct, which integrates structural, sequence and biochemical modalities, helps mitigate the impact of potential inaccuracies in individual structures. Improved upstream structural accuracy could further enhance downstream immunogenicity prediction, and the modular design of ImmunoStruct allows the incorporation of more accurate structural data as better prediction tools become available.

The structure generation process was distributed across five machines, with each machine processing a subset of the total peptide sequences. On a single NVIDIA 2080 Ti GPU, AlphaFold2 generated 1 peptide–MHC structure in approximately 3 min. Using this distributed setup, the complete set of peptide–MHC complexes from the combined IEDB and CEDAR datasets was generated in approximately 36 h. This procedure scales efficiently with the number of available machines, and all generated structures are included in our release so that future users do not need to repeat this process.

Biochemical properties. Recognizing the potential benefit of including biochemical properties as additional features in our prediction task, we integrated biochemical properties into the ImmunoStruct prediction. The `peptides` package³² was used to calculate all possible 84 biochemical properties for each peptide. The top ten biochemical properties with the highest standard deviation were selected in addition to the peptide–MHC solvent-accessible surface-area score, computed as previously described in ref. 64. These biochemical and biostructural properties were then grouped into two meta-property features (Bchem1 and Bchem2) during training.

Peptide–MHC in vitro and clinical validation. IFN γ -based enzyme-linked immunospot (ELISpot) assays were used to validate a set of SARS-CoV-2 epitopes. ELISpot assays provide a standardized measurement of T cell responses^{65,66}. Previously frozen human peripheral blood mononuclear cells (PBMCs) were thawed rapidly to 37 °C and incubated overnight (Millipore). PBMCs were plated at 400,000 cells per well and co-incubated with 10 $\mu\text{g ml}^{-1}$ of peptide for 48 h at 37 °C. ELISpot plates were analysed using CTL Immunospot. Positive controls were PBMCs plated at 200,000 cells per well and co-incubated with Spike

Glycoprotein peptide megapool at $1 \mu\text{g ml}^{-1}$ (JPT Peptide Technologies). Negative controls were unstimulated PBMCs plated at 200,000 cells per well. Positive T cell responses were defined by greater or equal to the mean of negative controls +3. Three additional true negative peptide–MHC samples validated through similar methods were identified through literature review to improve data balance⁶⁷.

The clinical cohort comprised 70 patients with cancer undergoing immunotherapy. Wild-type and mutant peptides were curated as previously described in CEDAR^{20,46}. Each unique peptide–MHC pair was treated as an independent unique sample. For a patient where the same peptide was paired to different alleles, it was counted separately owing to peptide–MHC–TCR reactivity specificities. When the same peptide–MHC pair had alternative experimental labels, it was considered an immunogenic sample 1 if at least one ground-truth label was immunogenic. To ensure a uniform, no-retraining evaluation, predictions were run only on peptide–MHCs involving the 27 alleles on which the model was trained. Progression-free survival was defined as time from treatment initiation to radiographic or clinical progression (RECIST v.1.1).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The infectious disease data were obtained from IEDB at <https://iedb.org/>. The cancer neoepitope data were obtained from CEDAR at <https://cedar.iedb.org/>. The cancer survival data were obtained as previously described⁴⁶. Data are freely available via GitHub at <https://github.com/KrishnaswamyLab/ImmunoStruct>.

Code availability

The source code for the ImmunoStruct model and inference scripts are available under an open-source license via GitHub at <https://github.com/KrishnaswamyLab/ImmunoStruct> and are available via Zenodo at <https://doi.org/10.5281/zenodo.17535443> (ref. 68).

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Author contributions

K.B.G., A.I. and S.K. identified the research problem and designed this work. K.B.G. collected and cleaned the IEDB and CEDAR data. K.B.G., J.F.R., C.L., E.Y., R.Y., A.I. and S.K. conceived the experiments. K.B.G., J.F.R., C.L. and E.Y. developed ImmunoStruct. C.L. and

K.B.G. conceived, designed and developed the cancer–wild-type contrastive learning. K.B.G., J.F.R., C.L. and E.Y. trained and evaluated ImmunoStruct. K.B.G., K.G. and E.C. performed the experimental validation on SARS-CoV-2. K.G. and J.F.R. performed the clinical validation. K.B.G. and S.T. conducted the peptide–TCR contact analysis. K.B.G., J.F.R. and C.L. performed the data analysis. K.B.G., C.L. and S.T. produced the figures. All authors participated in the discussion and wrote the paper.

Competing interests

S.K. is the chief scientific officer of Latent Alpha and cofounder of Ascent Bio. A.I. is a member of the board of directors of Roche Holding Ltd and of Genentech. A.I. co-founded RIGImmune, Xanadu Bio and PanV. R.Y. is a Amazon Scholar. E.C. is cofounder of Neomabs Biotechnologies Inc. The other authors declare no competing interests.

Additional information

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Replication Models were retrained from scratch using the provided scripts (<https://github.com/KrishnaswamyLab/ImmunoStruct/tree/main>) to obtained replicated results. Multiple team members obtained same results independently.

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